Point-by-Point Author Response to Reviewers

Manuscript ID: IDCIoT-154

Title: CoT-HMM: Structural Plausibility Scoring of Chain-of-Thought Reasoning via Hidden Markov Models

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We sincerely thank the Technical Program Committee and both anonymous reviewers for their constructive, rigorous, and encouraging feedback on our manuscript. We have thoroughly revised the paper according to your recommendations. Below is our detailed point-by-point response addressing each remark, with direct references to the revised sections of the manuscript.

Response to Reviewer 1

Remark R1.1: *Evaluates only 39 correct traces from Qwen2.5-0.5B; sensitive sample size.*

- **Response:** In Section IV-A (*Setup: Baselines and Metrics*), we have explicitly clarified that the $N = 39$ answer-verified correct traces from Qwen2.5-0.5B (and their 4 derived corruption variants, $N = 195$ total variants) serve as a focused pilot perturbation benchmark. We explicitly acknowledge that variants derived from the same source problem share underlying problem complexity.

- To account for sampling variance, non-parametric percentile bootstrap resampling ($B =$ 10,000 iterations) was used to compute robust 95% confidence intervals across all reported metrics (Table II).

- Furthermore, in Section IV-C and Table IV (*Zero-Shot Cross-Architecture Transfer*), we evaluate cross-model transfer on three distinct architectures without retraining: Llama-3.2-1B ($r = 0.946$), Phi-3-mini ($r = 0.941$), and Mistral-7B ($r = 0.970$) (all $p < 0.001$), supporting the consistency of structural transition regularities.

Remark R1.2: *Synthetic corruption vs. natural reasoning failures.*

- **Response:** Controlled perturbation (step deletion, loop injection, premature truncation) enables causal isolation of structural failure modes without confounding noise from arithmetic mistakes or lexical vocabulary shifts, an established methodology in reasoning evaluation benchmarks (e.g., ROSCOE).

- Crucially, this synthetic detection directly mirrors real-world reasoning failures: in uncorrupted, organic model generations on GSM8K (Section IV-D), **77.8% of looping chains ($N = 150$) were mathematically incorrect vs. 23.8% of non-looping ones** ($z = 6.18, p = 6.3 \times 10^{-10}$).

Remark R1.3: *Regex-based semantic tagger ambiguity & error propagation.*

- **Response:** In Section III-B, we report that our rule-based tagger achieved Cohen's $\kappa = 0.953$ agreement against a 50-step expert gold standard (comparable to Phi-3-mini's $\kappa = 0.976$).

- We expanded the dedicated paragraph (*Tagger Error Analysis*) in Section III-B detailing failure modes: arithmetic notation in problem restatements, connective adverbs ("thus", "hence") in intermediate calculations, and intermediate values resembling conclusions. Because the Ensemble HMM models continuous emission distributions across $M = 12$ models, isolated tagging fluctuations are softly absorbed across the full sequence. In Section VI, fine-tuning a lightweight DeBERTa-v3 sentence classifier is proposed as a direct future extension.

Remark R1.4: *Table II AUC: 0.658 for missing computation & 0.761 for circular reasoning vs. high for premature conclusions.*

- **Response:** In Section IV-B, we added an explanatory paragraph (*Asymmetric Detection Dynamics*) candidly assessing these boundaries:
  - *Premature Conclusions ($AUC = 0.990$, 95% CI [0.966, 1.000]):* Directly jumping from problem restatement to conclusion (PR $\rightarrow$ CO) represents a severe terminal transition violation ($P(\text{CO} \mid \text{PR}) < 10^{-4}$ in the learned matrix), triggering a catastrophic log-likelihood penalty ($d = 4.166$).
  - *Circular Reasoning ($AUC = 0.761$, 95% CI [0.645, 0.866]):* Moderate discrimination ($d = 0.896$); repeated loops trigger cumulative transition penalties, though legitimate multi-step deductions occasionally overlap.
  - *Missing Computation ($AUC = 0.658$, 95% CI [0.531, 0.778]):* Omitting arithmetic calculation tags (CA) leaves residual sequences of logical deductions (PR $\rightarrow$ IN $\rightarrow$ IN $\rightarrow$ CO) that locally resemble valid qualitative reasoning. We candidly note that the lower bound approaches chance (0.50). Token-length weighting and entropy thresholds provide immediate avenues to boost missing-step sensitivity.

Remark R1.5: *Distinction between Table II (0.990) and Table III (0.982) AUC.*

- **Response:** Table II reports *per-category controlled error injection AUCs* on the pilot corpus to isolate specific failure modes. Table III reports the *composite out-of-fold plausibility discrimination AUC* (0.982) on mixed benchmark traces against baseline verifiers (ROSCOE, PiCSAR, Length, Random). We added clear cross-referencing in Section IV-B to clarify this distinction.

Remark R1.6: *Conclusion should be a separate section after Results.*

- **Response:** We restructured the manuscript to separate Discussion and Conclusion. Section V is now dedicated to **Discussion**, and Section VI is a standalone section titled **Conclusion and Future Work**.

Remark R1.7: *Proofread for grammar/spelling/typos.*

- **Response:** A thorough editorial pass was performed across all sections:
  - Corrected Section III-D: Replaced the misnomer "Viterbi training" with "Expectation-Maximization (Baum-Welch) training", clarifying that Viterbi decoding is used to compute optimal hidden state paths.
  - Clarified training data as "answer-verified" (matching benchmark ground truth).
  - Refined equation notation, float spacing, and caption clarity.

Response to Reviewer 2

Remark R2.1: *How does CoT-HMM handle arithmetic errors?*

- **Response:** In Section V, we added a dedicated subsection titled **Arithmetic Correctness & Scope**. CoT-HMM evaluates reasoning grammar, not numerical arithmetic. An equation such as $12 \times 4 = 52$ possesses valid reasoning syntax (EQ $\rightarrow$ CA $\rightarrow$ IN); hence, CoT-HMM appropriately scores it as structurally plausible.
- This is empirically demonstrated in Table II, where our semantic-only corruption condition (pristine structure, incorrect arithmetic) yielded AUC = 0.501 (random guessing). Detecting numerical errors belongs to downstream deterministic tools (e.g., Python REPL / calculators). CoT-HMM acts as a fast ($\approx$150ms), zero-GPU filter that eliminates structurally broken chains before invoking heavier tools.

Remark R2.2: *Missing-computation detection AUC remains insufficient (0.658).*

- **Response:** We concur with the reviewer and candidly acknowledge this limitation. As explained in Remark R1.4, omitting calculation steps preserves valid qualitative deduction sub-chains (PR $\rightarrow$ IN $\rightarrow$ CO), resulting in AUC = 0.658 [0.531, 0.778]. In Section IV-B and Section VI, we explicitly discuss this boundary and outline mitigations via token-length weighted transition matrices and posterior Structural Entropy thresholds.

Remark R2.3: *How was the 99.3% routing accuracy calculated?*

- **Response:** In Section IV-D, we have clarified that the 99.3% metric represents conditional precision within the high-confidence pass tier, formulated as:

$$\text{Prec}_{\text{pass}} = \frac{|\mathcal{T}_{\text{pass}} \cap \mathcal{C}_{\text{correct}}|}{|\mathcal{T}_{\text{pass}}|} = \frac{TP}{TP + FP} = 99.3\% \tag{1}$$

  where $\mathcal{T}_{\text{pass}}$ contains traces with plausibility margin exceeding the decision threshold τ_{high} (calibrated via Youden's J on validation data), and $\mathcal{C}_{\text{correct}}$ denotes ground-truth correct chains. Traces in this top 35% tier bypass secondary verification with 99.3% precision. Empirical token savings (42%–47%) are evaluated relative to invoking secondary verifiers across all candidate traces.

Remark R2.4: *Regex-based semantic tagging needs error analysis.*

- **Response:** Addressed in Section III-B (*Tagger Error Analysis*), outlining failure modes (overlapping discourse markers, conversational cues) and explaining why the Ensemble HMM's soft emission distributions prevent individual misclassifications from corrupting sequence-level plausibility scores.

Remark R2.5: *Semantic error detection limitations require discussion.*

- **Response:** In Section V (*Discussion*), under **Arithmetic Correctness & Scope** and **Limitations**, we expand on the complementary division of labor: CoT-HMM rapidly detects macroscopic structural breakdowns (loops, premature stops, disordered logic), while specialized semantic verifiers (e.g., PRMs, Python REPL) verify numerical and factual correctness, reducing total token overhead by 42%–47%.

Summary of Manuscript Modifications (IEEE IDCIoT-154)

Section	**Revisions Incorporated**
Section III-B	Added tagger priority order, error analysis, and context-dependent boundary cases.
Section III-D	Corrected "Viterbi training" to Expectation-Maximization (Baum-Welch) training.
Section IV-A	Clarified $N = 39$ pilot corpus, $N = 195$ variants, bootstrap CIs, and multi-model transfer.
Section IV-B	Candid assessment of AUCs (0.990, 0.761, 0.658, 0.501); Table II vs. Table III distinction.
Section IV-D	Added Eq. (3) defining 99.3% pass-tier routing precision and empirical token savings.
Section V	Dedicated Discussion section analyzing structural vs. arithmetic correctness and scope.
Section VI	Standalone *Conclusion and Future Work* section outlining concrete research directions.
Throughout	Complete proofreading, float tuning, and verified exact 5-page IEEEtran compliance.